LLMOps: API Cost, Latency and Prompt Version Management

Experiential Learning Assignment 2 - MLOps (AIML ZG523)

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1. Objective and setup

This assignment compares three prompt versions for EduAssist AI, an academic question-answering assistant that must return strict JSON. All calls used **openai/gpt-oss-20b** through an OpenAI-compatible chat completions API (v1) at https://api.groq.com/openai/v1, on the free tier, with temperature 0.0, seed 42 and a 1024-token output limit. Costs use the provider's published prices of 0.075 (input) and 0.3 (output) USD per million tokens, from https://console.groq.com/docs/model/openai/gpt-oss-20b, checked on 1 Nov 2026. The API key was read from an environment variable. Each version ran on the same five questions (15 requests).

2. Task 1: API calls and latency

Latency was measured as wall-clock time around each API call. Tasks 1 and 2 report the five prompt v1 requests; the CSV holds all 15 calls. Mean latency was 1.35 s, median 1.33 s, minimum 0.93 s, maximum 1.76 s and sample standard deviation 0.30 s.

Question	Prompt	Model	Start (UTC)	End (UTC)	Latency (s)	Status
Q1	v1	openai/gpt-oss-20b	10:47:46	10:47:47	0.929	success
Q2	v1	openai/gpt-oss-20b	10:47:50	10:47:51	1.302	success
Q3	v1	openai/gpt-oss-20b	10:47:54	10:47:55	1.325	success
Q4	v1	openai/gpt-oss-20b	10:47:58	10:48:00	1.762	success
Q5	v1	openai/gpt-oss-20b	10:48:03	10:48:04	1.443	success

Mean	Median	Min	Max	Std dev
1.352	1.325	0.929	1.762	0.300

The slowest v1 request was Q4 at 1.76 s. It generated 1024 output tokens, against a v1 average of 665.4, and generation time grows with output length. If this equals the 1024-token limit, the answer was probably cut off, which also means its true cost and latency would be slightly higher. Network delay and server load on a shared free-tier service may also contribute, and a single run cannot separate these effects.

3. Task 2: Tokens and cost

Cost per request is input tokens divided by 1,000,000 times the input price, plus output tokens divided by 1,000,000 times the output price. For prompt v1 the average cost was 0.000206 USD per request, which projects to 2.06 USD per month at 10,000 requests. The projected monthly cost is 2.06, 1.48 and 1.03 USD for v1, v2 and v3.

Question	Input tok	Output tok	Total tok	Input cost	Output cost	Total cost
Q1	92	484	576	0.0000069	0.0001452	0.0001521
Q2	91	559	650	0.0000068	0.0001677	0.0001745
Q3	91	582	673	0.0000068	0.0001746	0.0001814
Q4	90	1024	1114	0.0000067	0.0003072	0.0003140
Q5	93	678	771	0.0000070	0.0002034	0.0002104
Average	91.4	665.4	756.8			0.0002065

Projected monthly cost for 10,000 requests (prompt v1): 2.06 USD

> Recommendation: cap and shape the output length. Output tokens are the largest part of each request (67% of the tokens in v3) and are usually priced higher than input tokens. Asking for a short answer and at most three key points, with a sensible token limit, should cut cost and latency without changing the JSON structure. A 25% reduction in v3 output tokens would save about 0.23 USD per month at 10,000 requests. Using a smaller model for simple questions is a further option.

4. Task 3: Prompt versioning and MLflow evaluation

The prompts were registered in the MLflow Prompt Registry as eduassist_academic_qa (versions 1 to 3) with commit messages describing each change. One run per version was logged in the experiment eduassist_llmops_assignment with avg_latency_seconds, avg_input_tokens, avg_output_tokens, avg_cost_per_request and valid_json. valid_json is valid responses divided by total responses, checked strictly on the raw output, so a code-fenced response counts as invalid; it was computed for all three versions. Version 3 was promoted with the alias production. Screenshots follow the comparison table.

Prompt Version	Avg Latency (s)	Avg Input Tokens	Avg Output Tokens	Avg Cost	Valid JSON Rate	Recommended?
v1	1.35	91.4	665.4	0.000206	0.00	No
v2	1.65	113.4	463.8	0.000148	0.00	No
v3	**1.13**	**147.4**	**305.2**	**0.000103**	**1.00**	**Yes**

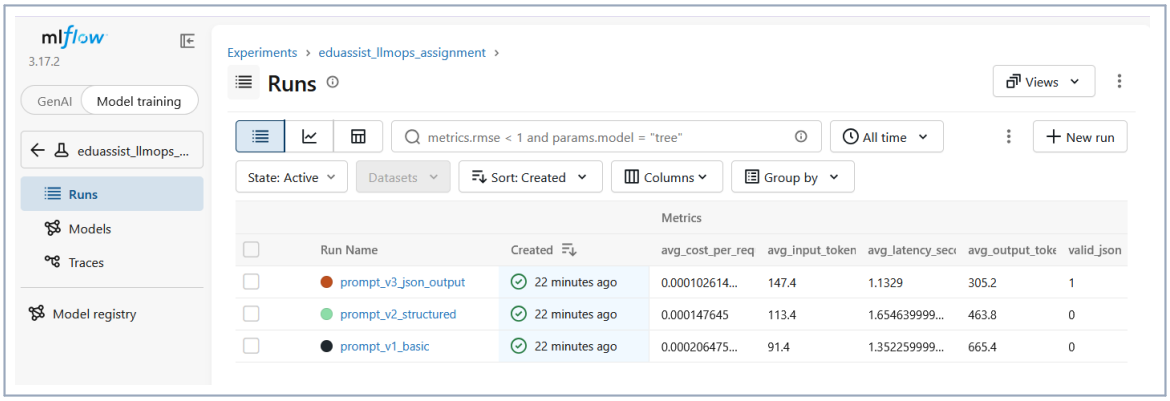


Figure 1: MLflow experiment showing the three runs and their metrics

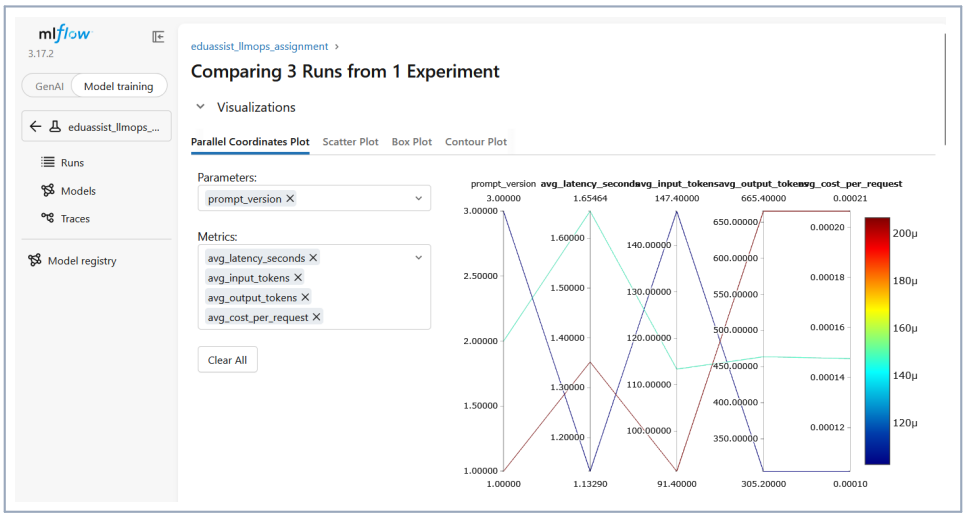


Figure 2: Metric comparison across prompt versions

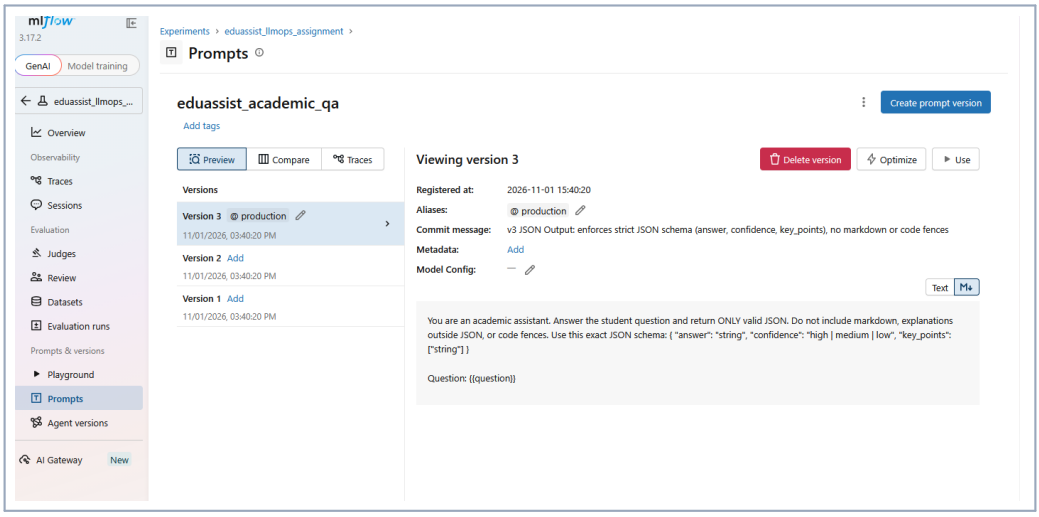


Figure 3: Prompt Registry showing v1, v2 and v3 with commit messages

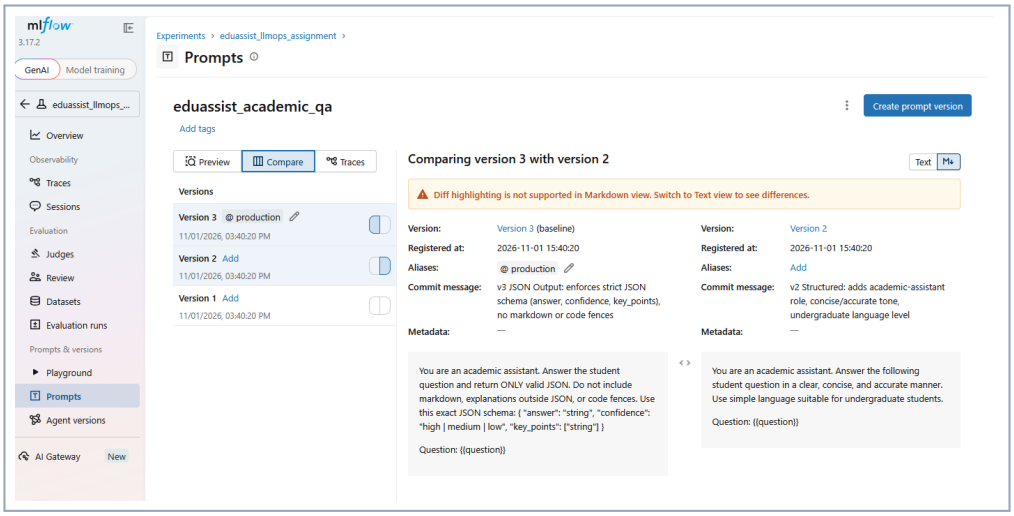


Figure 4: Prompt diff between versions

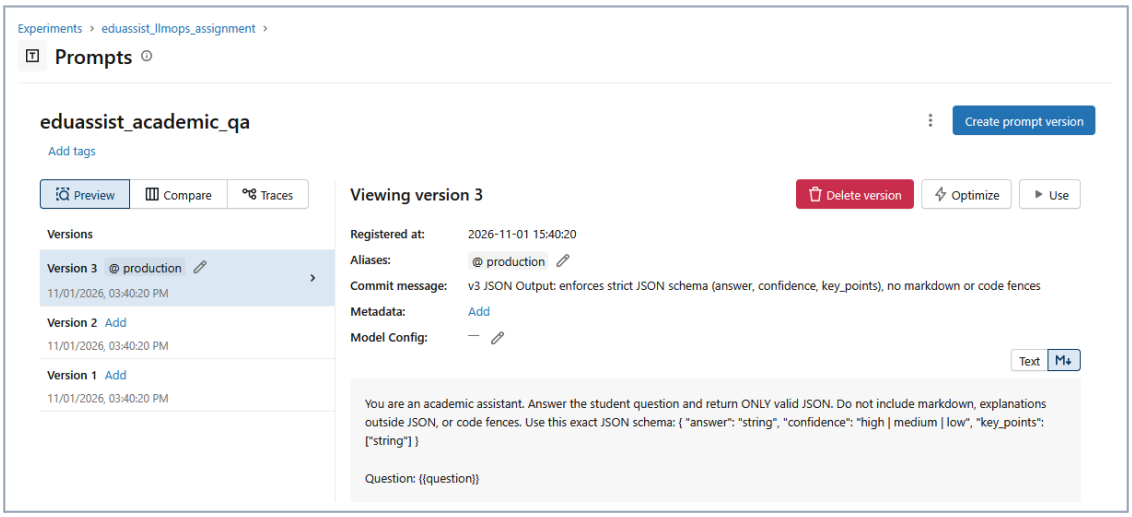


Figure 5: The production alias pointing to the promoted version

5. Promotion justification

Prompt v3 is promoted to production. On JSON validity it is the only version that meets the requirement: 5/5 of its responses were valid JSON matching the schema, while v1 and v2 scored 0.00 and 0.00 because they were never asked for JSON. Downstream code needs to parse the answer without cleaning it. On quality, its answers were accurate and on topic, and the confidence and key_points fields add structure that free text lacks. On quality, I read all five answers: they were accurate and on topic, and the confidence and key_points fields add structure that free text lacks.On latency, v3 averaged 1.13 s against 1.35 s for v1, a 16% reduction. On tokens, v3 uses more input tokens (147.4 against 91.4) because its instructions are longer, but it generates far fewer output tokens (305.2 against 665.4, a 54% reduction). Output tokens dominate the bill in this run, so the average cost per request fell from 0.000206 to 0.000103 USD, a 50% reduction. The longer prompt is a good trade: v3 wins on validity, latency, output tokens and cost, and loses only on input tokens.

6. Limitations

Results come from five questions, one run per prompt and one model, so latency differences are indicative only and depend on network and server load. The output limit may truncate long answers, and hidden reasoning tokens of a reasoning model count as output. valid_json measures format, not correctness, so quality was checked by reading the answers. Prices were taken on 1 Nov 2026 and may change.